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https://github.com/antoniomachuca/conway-99-graph

Date: October 3, 2026

To:
The Editors-in-Chief and Editorial Board
Journal of Automated Reasoning
Springer

Subject: Submission of the manuscript "Symmetry-Restricted SAT Encodings, Lean 4 Formalization, and Verification Limits for Conway's 99-Graph"

Dear Editors,

Please find enclosed the manuscript entitled **"Symmetry-Restricted SAT Encodings, Lean 4 Formalization, and Verification Limits for Conway's 99-Graph"**, which I submit for consideration for publication in the *Journal of Automated Reasoning*.

The manuscript concerns the existence of a strongly regular graph with parameters $\mathrm{srg}(99, 14, 1, 2)$, an open problem on J. H. Conway's list of five \$1,000 problems. It combines interactive theorem proving in Lean 4 with SAT solving and DRAT proof checking, and it states for each claim which of these checks the claim has passed. It does not decide existence or rigidity. The contents are:

1. **Kernel-checked graph formalization (Lean 4).** Proofs of K_4-freeness, the $7K_2$ decomposition of every neighborhood, diameter at most 2, and soundness and completeness of a computable checker for the adjacency predicate; the kernel reports only the axioms `[propext, Quot.sound]`. Arithmetic contradictions from character-theoretic arguments are also kernel-checked; wrappers that assume their premises are COMPILED.
2. **A kernel-checked order-7 reduction.** From a Conway 99-graph and a nontrivial automorphism s with $s^7 = \mathrm{id}$, a Lean proof constructs a canonical 15×15 orbit quotient matrix. It derives the fixed vertex, the orbits, and the quotient equations; it does not assume them. The kernel reports only `[propext, Quot.sound]`. Lean does not prove that no such matrix exists, so the Lean order-7 statement still depends on `sorryAx`. No proved link connects the matrix to a CNF.
3. **A checker-verified order-7 CNF refutation.** For one canonical order-7 CNF with Crawford symmetry-breaking predicates (177,097 variables, 422,508 clauses), CaDiCaL 1.9.5 returns `s UNSATISFIABLE`, and `drat-trim` reports `s VERIFIED` by backward checking of the 26,746,488,053-byte DRAT proof (no RAT lemmas in the core). Behbahani and Lam (2011) already excluded order 7. The graph-to-CNF translation is not kernel-verified, so the manuscript claims no new proof of that exclusion.
4. **Exterior-row CNF, separate from the Lean reduction.** The exterior-row CNF has 26,166 variables and 311,703 clauses. It is not the canonical order-7 CNF, and it is not in the Zenodo record. CaDiCaL 1.9.5 returned `s UNSATISFIABLE`, and `drat-trim` reported `s VERIFIED`. That refutation is PROVED for this CNF only. The Lean derivation of the exterior equations is a different PROVED component. The implication from `Z7OrbitMatrix` to the CNF is PENDING. The DRAT proof is not checked inside the Lean kernel, and Lean does not prove that no orbit matrix exists.
5. **Encoding validation.** A compiler defect treated variable index 1 as the constant `True`, so $x_1 + x_2 = 1$ became the unit clause $\neg x_2$. The manuscript describes the repair and the regression tests. Positive controls apply the production compilers for the involution with one fixed vertex and the fixed-point-free order-3 action to Paley(9) and to the Berlekamp–van Lint–Seidel graph $\mathrm{srg}(243, 22, 1, 2)$. For the second graph, the known adjacency, extended to the auxiliary variables, satisfies all emitted clauses (more than 27×10^6 per action). These controls do not prove compiler correctness, and they do not cover the order-7 compiler. A separate file has four tests of the fixed-3 encoding. The file is `tests/test_z3_fixed3_encoding_coverage.py`. The four tests passed. The four tests are COMPILED. The tests check 54 fixed one-factor entries. The tests check the circulant adjacency literal. The tests check 148 degree cardinalities. The tests check 1520 common-neighbor pair orbits. Those tests call no solver on the Conway fixed-3 formula. `conway_no_z3_fixed3_automorphism` still contains `sorry`. These tests do not cover the order-7 compiler. These tests do not cover the fixed-point-free order-3 encoding. Crnković and Maksimović (2020) already exclude this fixed-point case. These tests are not that exclusion.
6. **Open cases.** Every binary 7×42 matrix with $CC^T = 10I_7 + 2J_7$ has column weight 2 (involution with one fixed vertex). A 12.5-day CaDiCaL campaign (more than 2.4×10^9 conflicts, recorded aggregate) on the open involution and order-3 encodings found no model and no refutation; it started before a compiler repair.

Every claim carries one of four labels (PROVED, COMPILED, EXPLORED, PENDING) defined in the manuscript. The Lean 4 library, the SAT compilers, and the regression tests are available on GitHub (`https://github.com/antoniomachuca/conway-99-graph`). The canonical order-7 CNF and its DRAT proof are in Zenodo record 10.5281/zenodo.22995101. The solver log, the checker log, and the SHA-256 digests for that CNF are in the same record. The exterior-row CNF is not in that record.

Declarations: The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing financial or non-financial interests.

Suggested Reviewers: I suggest the following three referees:

1. **Prof. Marijn J. H. Heule** (Carnegie Mellon University, `marijn@cmu.edu`) — SAT solving and clausal proof checking; co-author of `drat-trim`.
2. **Prof. Armin Biere** (University of Freiburg, `biere@cs.uni-freiburg.de`) — Propositional satisfiability; lead developer of CaDiCaL.
3. **Prof. Jasmin Blanchette** (Ludwig-Maximilians-Universität München, `jasmin.blanchette@ifi.lmu.de`) — Automated reasoning, interactive theorem proving, proof certification.

Thank you for your consideration of this manuscript.

Sincerely,

Antonio Machuca